Practical - 5

Practical Title: Registration Form with Frontend Validation and User-Friendly Error Handling

Problem Definition: Create a student registration form with HTML5 input types and JavaScript validation for name, email, mobile number, password, confirm password, course, year, gender, and terms acceptance. Use Regular Expression for validations.

A. Theory

B. **1. Are correct input types and attributes used?**
Yes. The form uses appropriate input types such as `email`, `tel`, and `password`. Attributes such as `name`, `id`, and `value` are also used so that the form data can be identified and processed correctly. HTML input types also help browsers provide suitable validation and input controls. **Reference:** W3C – Validating Input

C. **2. Are validation errors displayed near the relevant fields?**
Yes. Validation errors are displayed near their respective fields using elements such as `nameError`, `emailError`, `mobileError`, and `passwordError`. This makes it easier for users to identify and correct the specific input that contains an error. **Reference:** W3C – Validating Input D. **3. Is password strength checked?**
Yes. The password is checked using a regular expression. It must contain at least 8 characters, including an uppercase letter, lowercase letter, number, and special character. This ensures that weak passwords are rejected during form validation.
Reference: W3C – Form Instructions

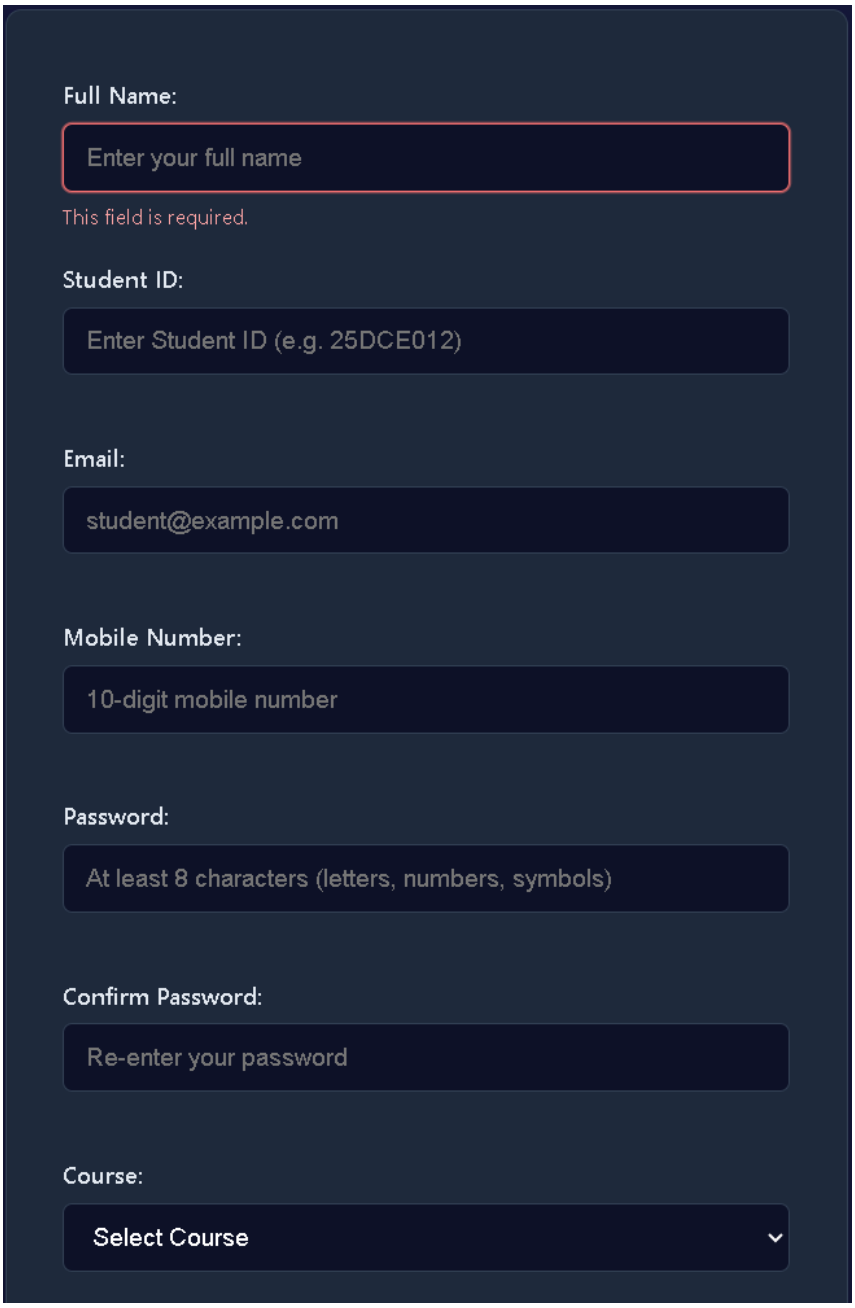
E. **4. Is the form accessible using keyboard and screen-reader-friendly labels?** The form uses standard HTML controls such as inputs, radio buttons, checkboxes, select menus, and a submit button, which can be operated using the keyboard. It also provides visible labels for the form fields. For better screen-reader accessibility, each `<label>` should be explicitly associated with its input using matching `for` and `id` attributes.
Reference: W3C – Labeling Controls

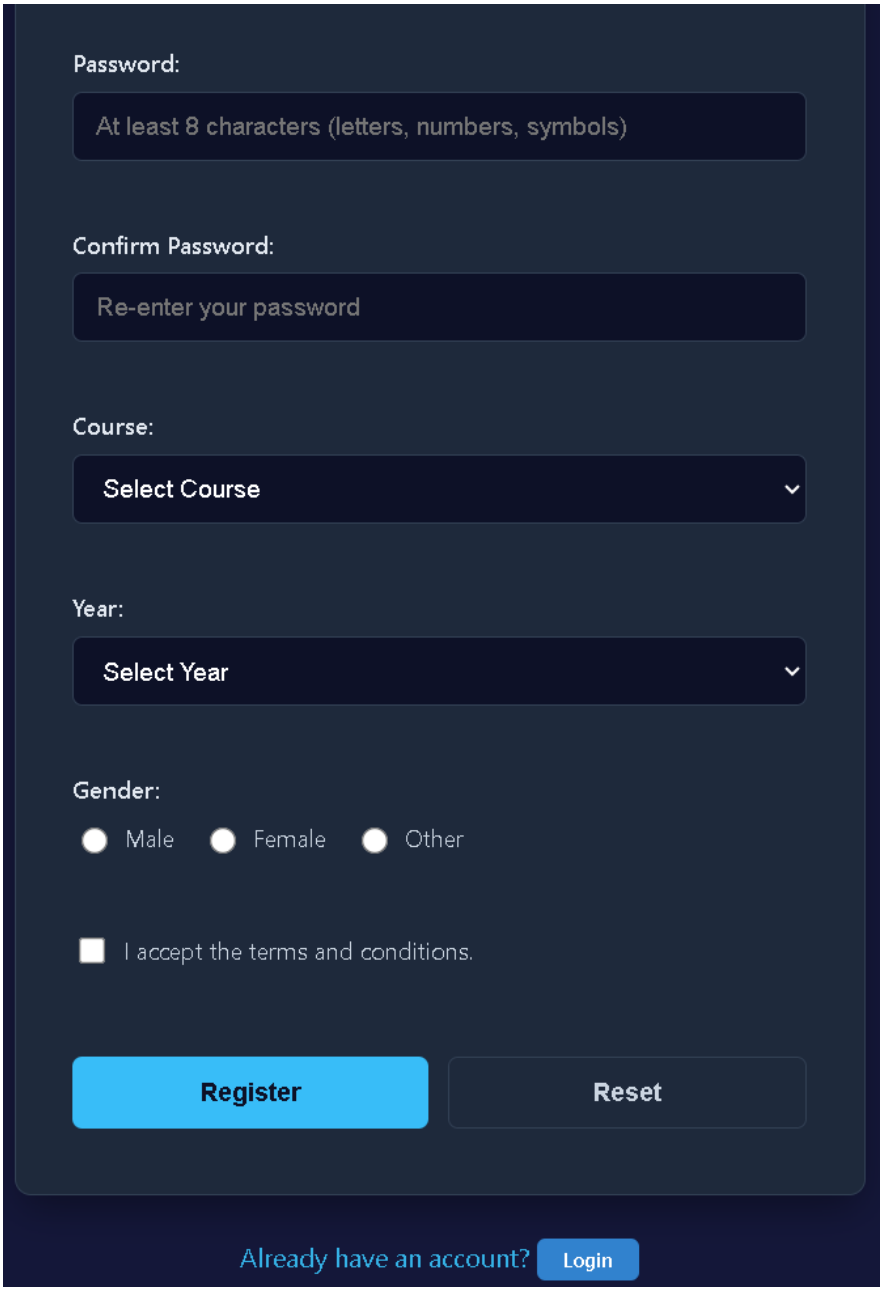
B. Implementation/ Output/Result

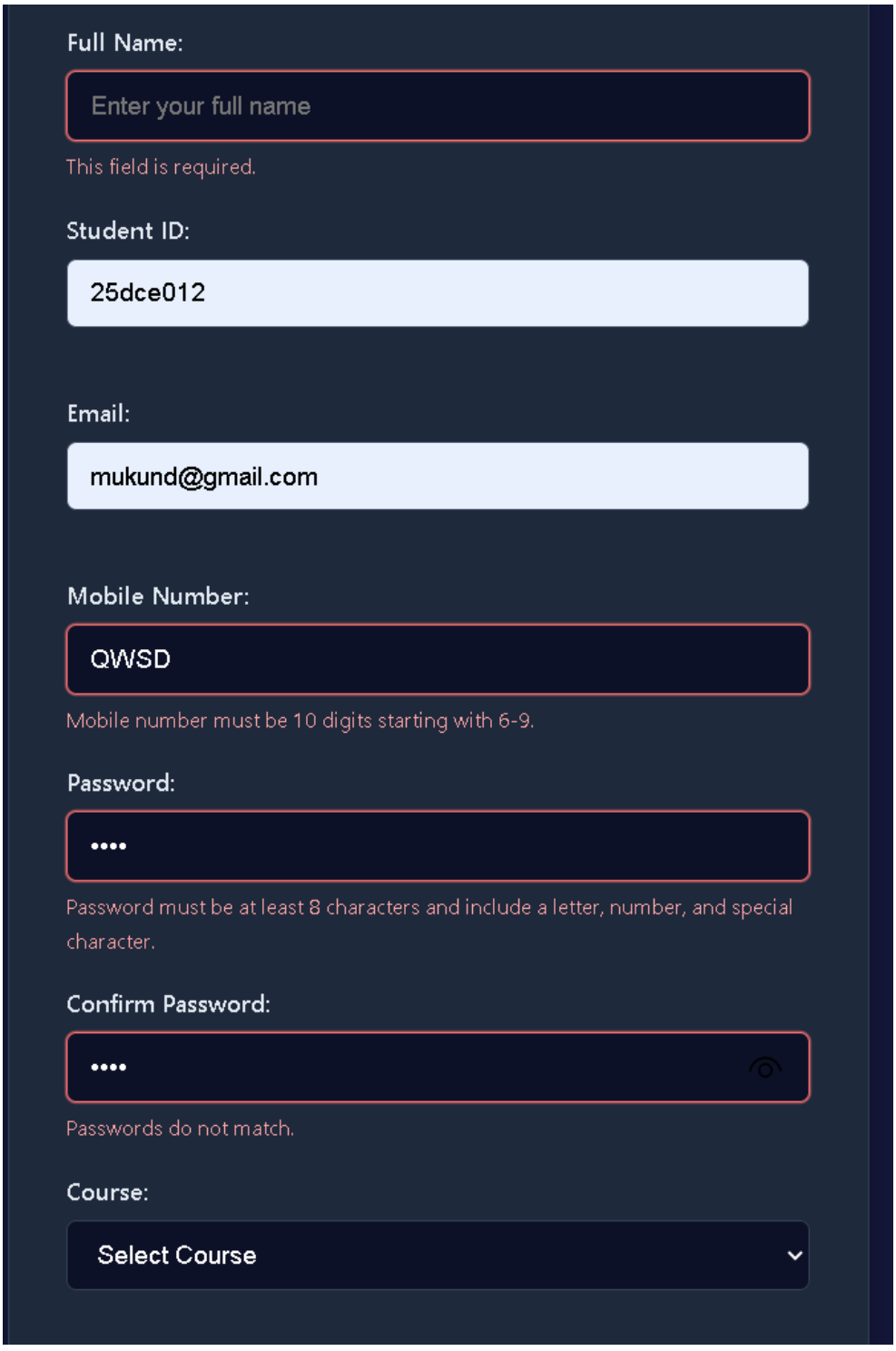
Result

The StudentHub registration page is now more accurate using javascript and drop down and other tags using regex in email and passwords to store proper data

C. Evidence of




Password:
At least 8 characters (letters, numbers, symbols)
Confirm Password:
Re-enter your password
Course:
Select Course
Year:
Select Year
Gender:
Male
Female
Other
I accept the terms and conditions.
Register
Reset
Already have an account?
Login


Full Name:
Enter your full name
This field is required.
Student ID:
25dce012
Email:
mukund@gmail.com
Mobile Number:
QWSD
Mobile number must be 10 digits starting with 6-9.
Password:
••••
Password must be at least 8 characters and include a letter, number, and special character.
Confirm Password:
••••
Passwords do not match.
Course:
Select Course

D. GitHub Repository Link:

GitHub Repository: https://github.com/justchill-spec

E. Challenges Faced:

In this practical the challenges I faced were the regex when I was entering the regex expression it still wasn't detecting it and the issue was the id name in html and javascript had a spelling difference mainly comparing the two inputs by user

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F. Learning Outcomes:

After this practical I learn how to use different javascript concepts and learned mainly how to use regex and comparing the two input given by the user like password and confirmpassword should be same